Problem Set 6

AI 501: Mathematics for Artificial Intelligence | Fall 2026

Problem 1

Consider a one-dimensional data set (x_n, y_n), $n = 1, \ldots, N$, and ridge regression with the intercept fixed at zero:

$$L(w) = \frac{1}{2}\sum_{n=1}^{N}(y_n - x_n w)^2 + \frac{\lambda}{2}\,w^2, \qquad \lambda \geq 0.$$

(a) Derive, step by step, a closed-form expression for the minimizer $w^\star$. Show that it is a minimizer and not just a stationary point.

(b) Intuitively, what effect does increasing λ have on the bias of $w^\star$? On its variance?

Solution:

(a) Differentiate and set the derivative to zero:

$$\frac{dL}{dw} = \sum_{n=1}^{N}\left(x_n^2 w - x_n y_n\right) + \lambda w = 0 \quad \Longrightarrow \quad w^\star = \frac{\sum_n x_n y_n}{\sum_n x_n^2 + \lambda}.$$

The second derivative is

$$\frac{d^2 L}{dw^2} = \sum_n x_n^2 + \lambda > 0$$

(for $\lambda > 0$, or for $\lambda = 0$ with at least one $x_n \neq 0$), so L is an upward-opening parabola and $w^\star$ is its minimum.

(b) Increasing λ pulls $w^\star$ towards 0, so the data have less influence on it: the *bias* increases and the *variance* decreases, since $w^\star$ fluctuates less from one data set to the next. Decreasing λ does the reverse.

Problem 2

Suppose $n = 2$ and $d = 2$, with

$$x_{11} = x_{12} = a, \qquad x_{21} = x_{22} = -a \quad (a \neq 0), \qquad y_1 + y_2 = 0.$$

Because the responses and each feature column sum to zero, the intercept estimate is zero and can be dropped. The ridge loss is

$$L(w) = \frac{1}{2}\sum_{i=1}^{2}\left(y_i - x_{i1}w_1 - x_{i2}w_2\right)^2 + \frac{\lambda}{2}\left(w_1^2 + w_2^2\right), \qquad \lambda > 0.$$

(a) Show that in this setting

$$L(w) = \big(y_1 - a(w_1 + w_2)\big)^2 + \frac{\lambda}{2}\left(w_1^2 + w_2^2\right).$$

(b) Show that the ridge coefficients satisfy $w_1^\star = w_2^\star$, and find their common value.

(c) Compute $w^\star$ for $a = 1$, $y_1 = 2$, $\lambda = 1$.

Solution:

(a) Write $s = w_1 + w_2$. The two residuals are

$$y_1 - aw_1 - aw_2 = y_1 - as, \qquad y_2 + aw_1 + aw_2 = -(y_1 - as),$$

using $y_2 = -y_1$. Both squares equal $(y_1 - as)^2$, so $\frac{1}{2} \cdot 2\,(y_1 - as)^2 = (y_1 - as)^2$, which gives the stated form.

(b) Setting both partial derivatives to zero:

$$\frac{\partial L}{\partial w_1} = -2a(y_1 - as) + \lambda w_1 = 0, \qquad \frac{\partial L}{\partial w_2} = -2a(y_1 - as) + \lambda w_2 = 0.$$

Subtracting gives $\lambda(w_1 - w_2) = 0$, and since $\lambda > 0$, $w_1 = w_2 = w$. Substituting $s = 2w$ into either equation:

$$-2ay_1 + 4a^2 w + \lambda w = 0 \quad \Longrightarrow \quad w_1^\star = w_2^\star = \frac{2ay_1}{4a^2 + \lambda}.$$

This stationary point is the unique minimizer, because the Hessian $\begin{pmatrix} 2a^2 + \lambda & 2a^2 \\ 2a^2 & 2a^2 + \lambda \end{pmatrix}$ has eigenvalues λ and $4a^2 + \lambda$, both positive.

(c) $w_1^\star = w_2^\star = \dfrac{2(1)(2)}{4 + 1} = 0.8$. Ridge splits the weight equally between the two identical features.

Problem 3

Let

$$A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}, \qquad A + I = \begin{pmatrix} 2 & 4 \\ 2 & 4 \end{pmatrix}.$$

(a) Find the eigenvalues and eigenvectors of A.

(b) Find the eigenvalues and eigenvectors of $A + I$. How are they related to those of A?

(c) What does one of the eigenvalues of $A + I$ tell you about the matrix $A + I$?

Solution:

(a)

$$\det(A - \lambda I) = (1 - \lambda)(3 - \lambda) - 8 = \lambda^2 - 4\lambda - 5 = (\lambda - 5)(\lambda + 1),$$

so $\lambda_1 = 5$ and $\lambda_2 = -1$.

- $\lambda_1 = 5$: $(A - 5I)q = \begin{pmatrix} -4 & 4 \\ 2 & -2 \end{pmatrix} q = 0$ gives $q_1 = (1, 1)^T$.
- $\lambda_2 = -1$: $(A + I)q = \begin{pmatrix} 2 & 4 \\ 2 & 4 \end{pmatrix} q = 0$ gives $q_2 = (-2, 1)^T$.

Check: $5 + (-1) = 4 = 1 + 3$ (the trace) and $5 \cdot (-1) = -5 = \det A$.

(b) $A + I$ has the *same* eigenvectors, with eigenvalues increased by 1: 6 and 0. In general, if $Aq = \lambda q$ then $(A + cI)q = (\lambda + c)q$.

(c) The eigenvalue 0 means $A + I$ is singular: $(A + I)(-2, 1)^T = 0$, its two rows are identical, and its columns are dependent (column 2 = 2× column 1). $A + I$ has no inverse.

Problem 4

A symmetric matrix acts as a pure scaling when points are described in coordinates along its eigenvectors. Every symmetric $A \in \mathbb{R}^{2\times 2}$ can be written $A = Q\Lambda Q^T$, where the columns q_1, q_2 of Q are orthonormal eigenvectors and $\Lambda = \mathrm{diag}(\lambda_1, \lambda_2)$. Reading $Ax = Q\Lambda Q^T x$ from right to left gives three steps:

(i) $u = Q^T x = (q_1^T x,\ q_2^T x)^T$: the coordinates of x along the eigenvectors;

(ii) $v = \Lambda u$: scale the two coordinates by λ_1 and λ_2;

(iii) $y = Qv = v_1 q_1 + v_2 q_2$: convert back to standard coordinates.

Use

$$A_1 = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} = \underbrace{\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}}_{Q} \underbrace{\begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}}_{\Lambda} \underbrace{\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}}_{Q^T},$$

$$A_2 = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix} = \underbrace{\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}}_{Q} \underbrace{\begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}}_{\Lambda} \underbrace{\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}}_{Q^T},$$

and the points $x_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $x_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $x_3 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Note that both matrices share the same Q.

(a) Write down q_1, q_2, λ_1, λ_2 for each matrix. Check that $Q^T Q = I$ and compute $\det Q$: is Q a rotation or a reflection?

(b) Compute $A_1 x$ and $A_2 x$ directly for each of the three points.

(c) Compute $u = Q^T x$ for each point (only once, since Q is shared). Which u has a zero entry, and why? Check that $\|u\| = \|x\|$; what does this say about how Q^T treats lengths?

(d) Compute $v = \Lambda u$ for each matrix. Along each eigenvector direction, is the point stretched or shrunk, by what factor, and is it flipped?

(e) Compute $y = Qv$. Record x, u, v, y in a table for each matrix, and confirm that each y matches part (b).

(f) Interpret your results.

(i) Which point is only scaled, never turned, by both matrices? Why?

(ii) Using the eigenvalues, explain why A_1 pulls x_1 and x_3 towards the diagonal.

(iii) Show that $A_2 = 2\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, and describe what A_2 does to the plane.

(g) Sketch the image of the unit circle under A_1 and under A_2, and relate its semi-axes to the eigenvectors and eigenvalues. Why is the flip in A_2 invisible in the shape of the image? *Hint: track where x_1 and x_3 go.*

Solution:

(a) For both matrices, $q_1 = \frac{1}{\sqrt{2}}(1,1)^T$ and $q_2 = \frac{1}{\sqrt{2}}(1,-1)^T$. For A_1: $\lambda_1 = 4$, $\lambda_2 = 2$. For A_2: $\lambda_1 = 2$, $\lambda_2 = -2$.

$$Q^T Q = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = I, \qquad \det Q = \frac{1}{2}\big((1)(-1) - (1)(1)\big) = -1.$$

So Q is a *reflection*: across the line through the origin at angle 22.5°, which swaps $e_1 \leftrightarrow q_1$ and $e_2 \leftrightarrow q_2$. Here Q is also symmetric, so $Q^T = Q$ and steps (i) and (iii) apply the same reflection. (Replacing q_2 by $-q_2$ gives the same A but makes Q a rotation by $+45°$; eigenvectors are only defined up to sign.)

(b)

$$A_1 x_1 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \quad A_1 x_2 = \begin{pmatrix} 4 \\ 4 \end{pmatrix}, \quad A_1 x_3 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}; \qquad A_2 x_1 = \begin{pmatrix} 0 \\ 2 \end{pmatrix}, \quad A_2 x_2 = \begin{pmatrix} 2 \\ 2 \end{pmatrix}, \quad A_2 x_3 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}.$$

(c)

$$u_1 = Q^T x_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad u_2 = Q^T x_2 = \begin{pmatrix} \sqrt{2} \\ 0 \end{pmatrix}, \qquad u_3 = Q^T x_3 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

- u_2 has a zero second entry: $x_2 = (1,1)^T$ lies along q_1, so it is perpendicular to q_2 and $q_2^T x_2 = 0$. It has no component along the second eigenvector.
- $\|u_1\| = \|x_1\| = 1$, $\|u_2\| = \|x_2\| = \sqrt{2}$, $\|u_3\| = \|x_3\| = 1$. The change of coordinates preserves lengths: it only re-describes x along new perpendicular axes, without stretching anything. Geometrically, Q^T carries q_1 onto the e_1 axis and q_2 onto the e_2 axis.

(d)
- A_1: $v_1 = \frac{1}{\sqrt{2}}(4,2)^T$, $v_2 = (4\sqrt{2},0)^T$, $v_3 = \frac{1}{\sqrt{2}}(4,-2)^T$. The q_1 direction is stretched by 4 and the q_2 direction by 2. Nothing is flipped, and since both $|\lambda_i| > 1$, nothing is shrunk.
- A_2: $v_1 = \frac{1}{\sqrt{2}}(2,-2)^T$, $v_2 = (2\sqrt{2},0)^T$, $v_3 = \frac{1}{\sqrt{2}}(2,2)^T$. The q_1 direction is stretched by 2. The q_2 direction is stretched by 2 *and flipped*, because $\lambda_2 = -2$.

This is the only step that changes shape, and each coordinate is scaled on its own, with no mixing.

(e) For example, for A_1 and x_1: $y = Qv = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}4\\2\end{pmatrix} = \frac{1}{2}\begin{pmatrix}6\\2\end{pmatrix} = \begin{pmatrix}3\\1\end{pmatrix}$.

A_1	u	v	y
x_1	$\frac{1}{\sqrt{2}}(1,\,1)^T$	$\frac{1}{\sqrt{2}}(4,\,2)^T$	$(3,\,1)^T$
x_2	$(\sqrt{2},\,0)^T$	$(4\sqrt{2},\,0)^T$	$(4,\,4)^T$
x_3	$\frac{1}{\sqrt{2}}(1,\,-1)^T$	$\frac{1}{\sqrt{2}}(4,\,-2)^T$	$(1,\,3)^T$

A_2	u	v	y
x_1	$\frac{1}{\sqrt{2}}(1,\,1)^T$	$\frac{1}{\sqrt{2}}(2,\,-2)^T$	$(0,\,2)^T$
x_2	$(\sqrt{2},\,0)^T$	$(2\sqrt{2},\,0)^T$	$(2,\,2)^T$
x_3	$\frac{1}{\sqrt{2}}(1,\,-1)^T$	$\frac{1}{\sqrt{2}}(2,\,2)^T$	$(2,\,0)^T$

Every y matches part (b). In one formula, $Ax = \lambda_1(q_1^T x)\,q_1 + \lambda_2(q_2^T x)\,q_2$: split x along the eigenvectors, scale each piece by its eigenvalue, and add.

(f) (i) $x_2 = (1,1)^T$. It lies along the eigenvector q_1, so u_2 has only a first coordinate, step (ii) only scales it, and step (iii) returns it to the same line: $A_1 x_2 = 4x_2$ and $A_2 x_2 = 2x_2$.

(ii) x_1 and x_3 have equal-sized components along q_1 and q_2. A_1 multiplies the q_1 (diagonal) component by 4 but the q_2 (anti-diagonal) component only by 2, so the diagonal part dominates the image: $(1,0) \mapsto (3,1)$ and $(0,1) \mapsto (1,3)$.

(iii) Directly, $2\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} = \begin{pmatrix}0 & 2\\ 2 & 0\end{pmatrix} = A_2$. A_2 reflects the plane across the diagonal line through $(1,1)$ (swapping the two coordinates) and doubles every length. This is exactly "stretch both eigen-directions by 2 and flip the q_2 direction".

(g) Write a point of the unit circle as $x = Qu$ with $\|u\| = 1$. Step (ii) turns the circle into an ellipse with semi-axes $|\lambda_1|$ and $|\lambda_2|$ along e_1 and e_2, and step (iii) turns those axes onto q_1 and q_2.

- A_1: an ellipse with semi-axis 4 along $(1,1)$ and semi-axis 2 along $(1,-1)$. Its area is $8\pi = |\det A_1|\,\pi$.
- A_2: a circle of radius 2, since $|\lambda_1| = |\lambda_2| = 2$. A circle looks the same whether or not it has been reflected, so the flip is invisible in the shape. It shows up in the orientation: x_1 at $0°$ maps to $90°$, while x_3 at $90°$ maps to $0°$. Walking counterclockwise around the input circle, the image moves clockwise, matching $\det A_2 = -4 < 0$.

Problem 5

This problem uses the eigenvalue decomposition to compute powers of a matrix, and links eigenvalues to the characteristic polynomial.

(a) Diagonalize

$$A = \begin{pmatrix}2 & -1\\ -1 & 2\end{pmatrix}$$

and compute $Q\Lambda^k Q^{-1}$ to prove that

$$A^k = \frac{1}{2}\begin{pmatrix}1+3^k & 1-3^k\\ 1-3^k & 1+3^k\end{pmatrix}.$$

(b) If a 2×2 matrix B has eigenvalues $\lambda_1 = 3$ and $\lambda_2 = 4$, what is $\det(B - \lambda I)$?

Solution:

(a) The characteristic polynomial is $(2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3)$.

- $\lambda_1 = 1$: $(A - I)q = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} q = 0$ gives $q_1 = \frac{1}{\sqrt{2}}(1,1)^T$.
- $\lambda_2 = 3$: $(A - 3I)q = \begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix} q = 0$ gives $q_2 = \frac{1}{\sqrt{2}}(-1,1)^T$.

Check: $A(1,1)^T = (1,1)^T$ and $A(-1,1)^T = (-3,3)^T = 3(-1,1)^T$. Since A is symmetric, the eigenvectors are orthonormal and $Q^{-1} = Q^T$:

$$Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \qquad \Lambda = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}.$$

Then

$$A^k = Q\Lambda^k Q^T = \frac{1}{2} \begin{pmatrix} 1 & -3^k \\ 1 & 3^k \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1+3^k & 1-3^k \\ 1-3^k & 1+3^k \end{pmatrix}.$$

Setting $k = 1$ recovers $\frac{1}{2} \begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix} = A$. Geometrically, A leaves the diagonal unchanged and stretches the anti-diagonal by 3, so k applications stretch the anti-diagonal by 3^k.

(b) $\det(B - \lambda I) = (3-\lambda)(4-\lambda) = \lambda^2 - 7\lambda + 12$. In general, the eigenvalues of a 2×2 matrix are the roots of $t^2 - (\operatorname{tr} B)\, t + \det B$. For the matrix A of part (a), with eigenvalues 1 and 3, this gives $\lambda^2 - 4\lambda + 3$, matching the characteristic polynomial found there.